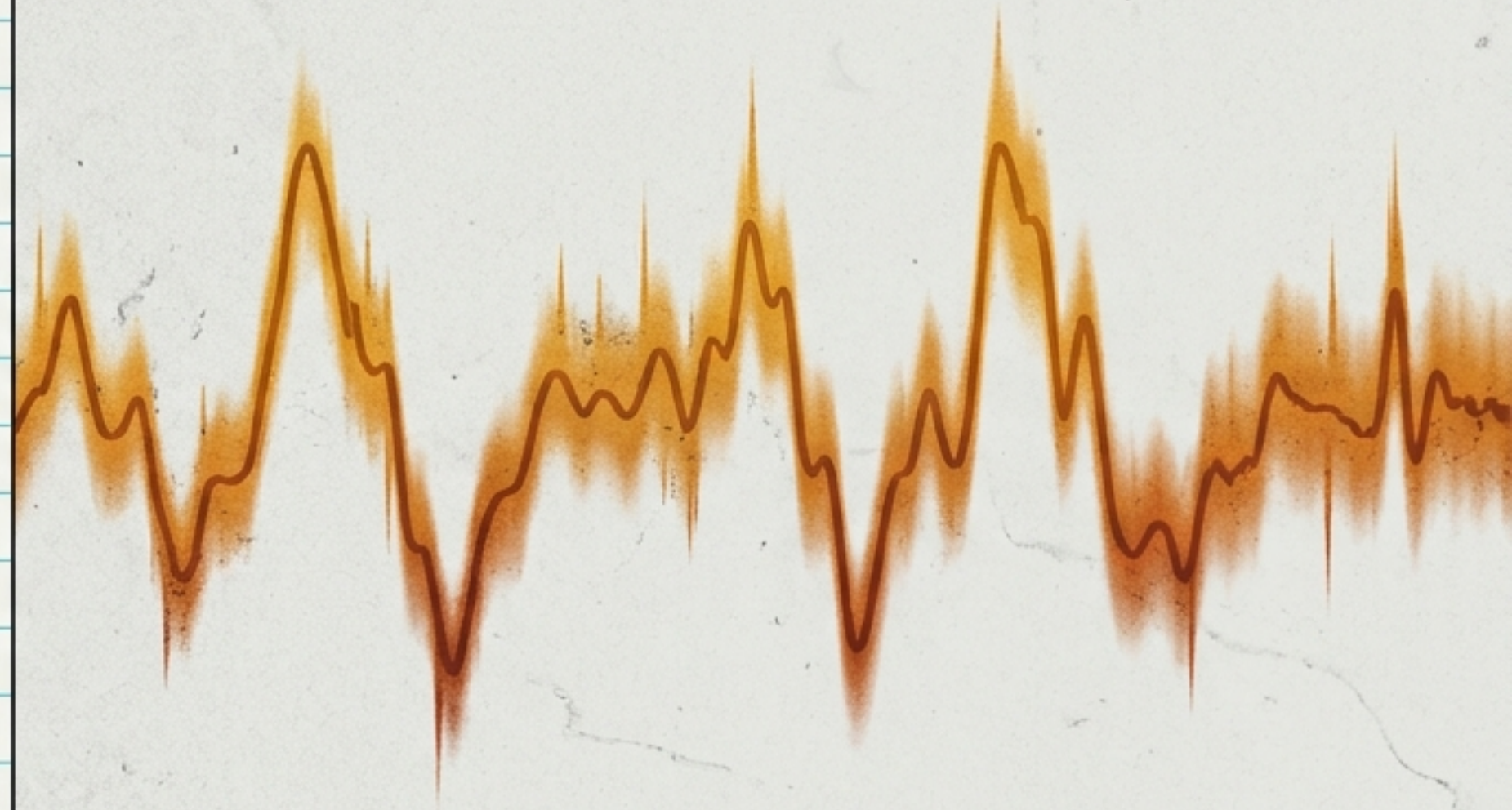


Signals, Noise, & Attenuation

```
1011010010110100  
0100101101001011  
1101001011010010
```

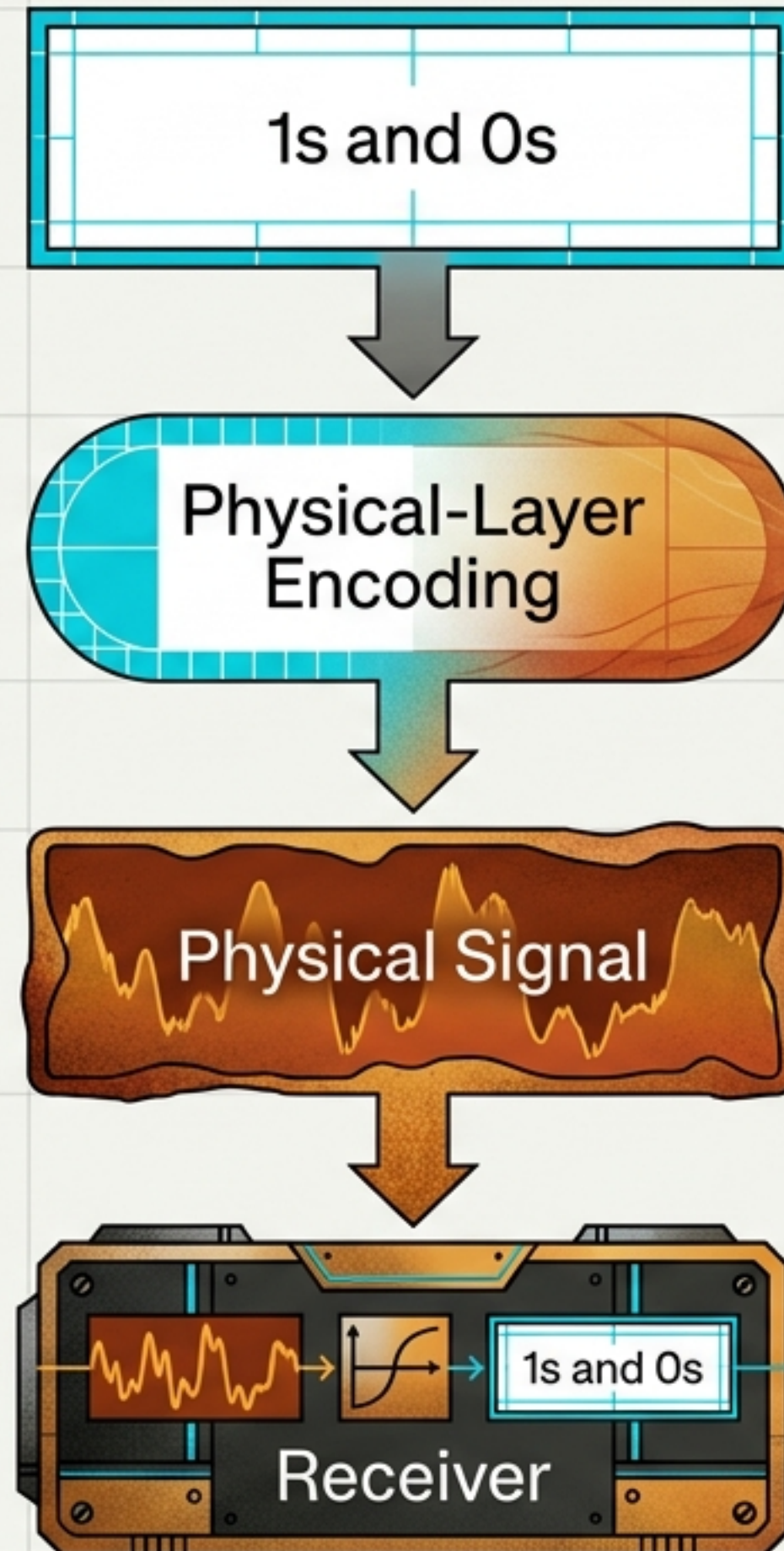
The Logical Domain
(Abstract Data)

The Physical Reality of Networking



The Physical Domain
(Real World)

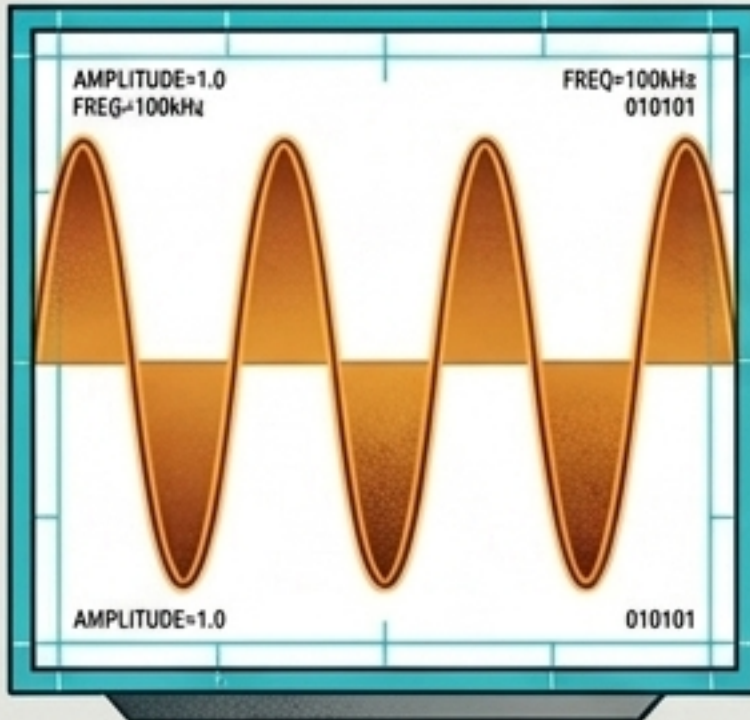
Abstract bits cannot travel through a cable. They must become physical energy. The physical layer forms the connection between the logical networking system and the real physical world.



Copper (Twisted-Pair)	Fiber Optic	Wireless (Wi-Fi)
		
- Physical Signal: Electrical Energy - Transmission Environment: Metallic conductors - Key Characteristic: Susceptible to electromagnetic interference (EMI)	- Physical Signal: Optical Energy (Light) - Transmission Environment: Glass cladding and core - Key Characteristic: Extremely long distance; immune to EMI	- Physical Signal: Radio-frequency electromagnetic signal - Transmission Environment: Free space - Key Characteristic: Uncontained medium; signals can be intercepted

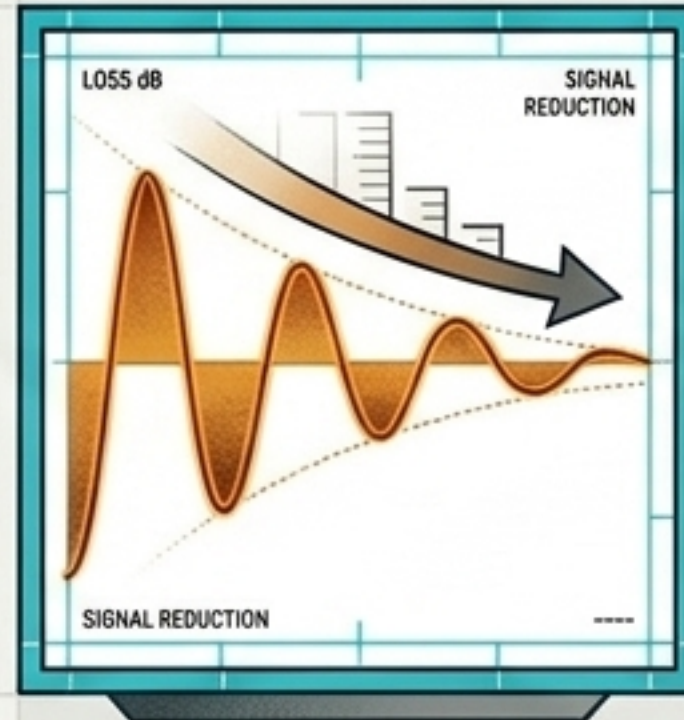
One message, different physical forms. The higher-level protocol (e.g., TCP/IP) does not care which medium is used.

[Transmitted Signal]



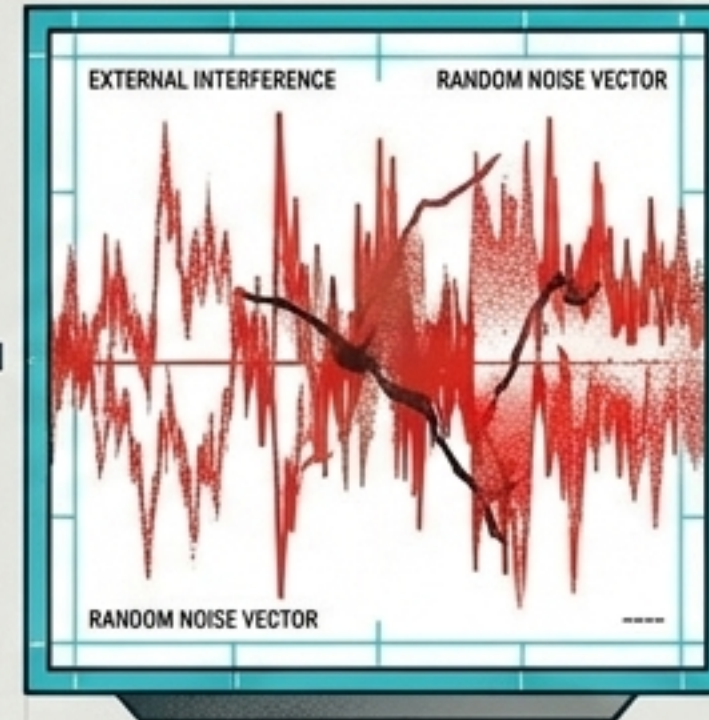
+

[Attenuation]



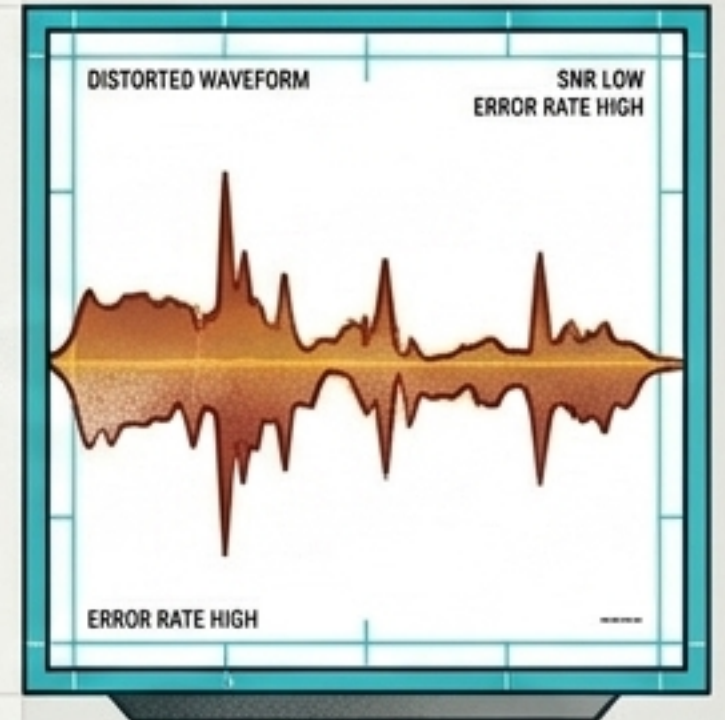
+

[Noise & Interference]



=

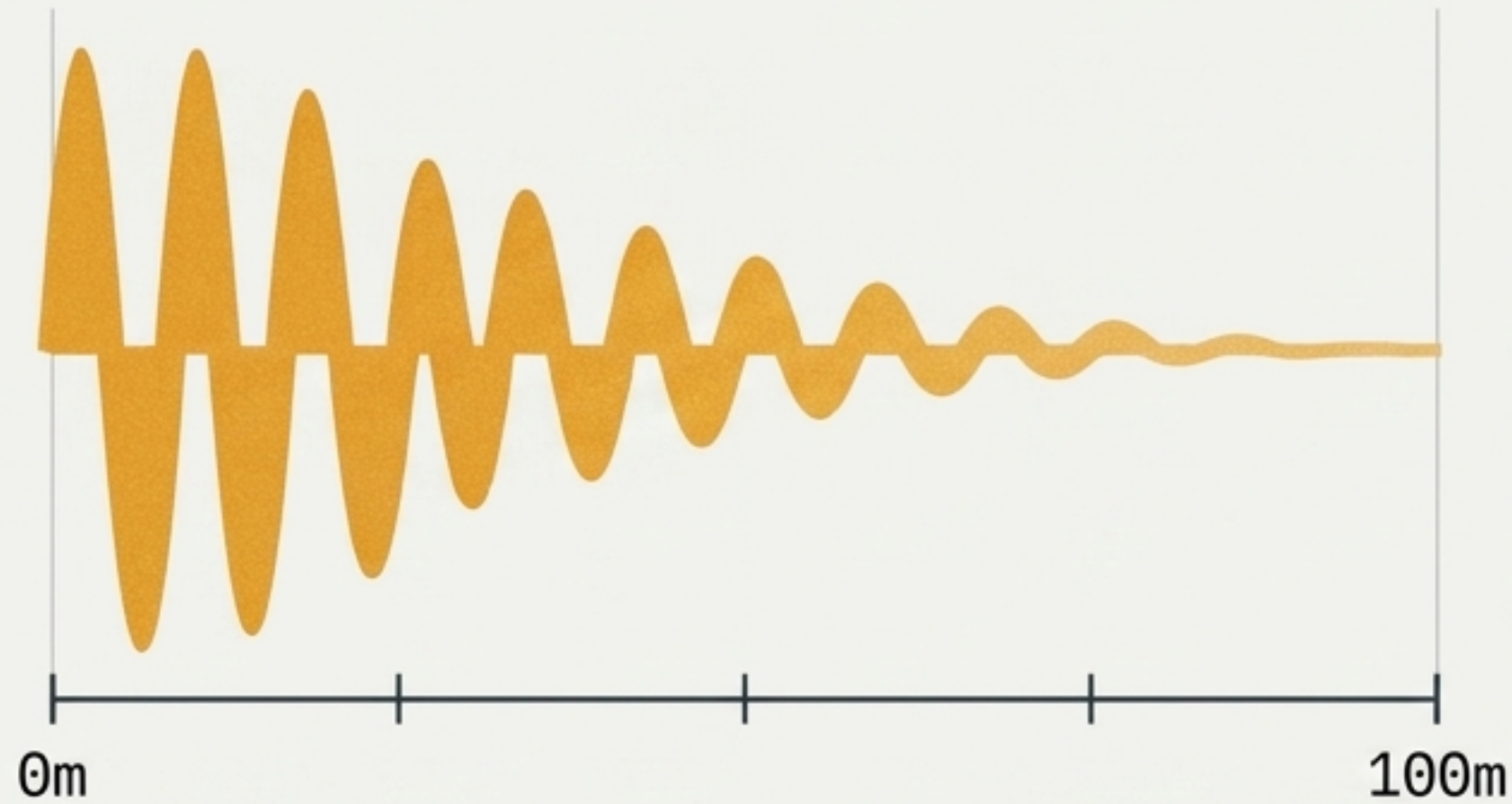
[Received Signal]



The Receiver's Dilemma

The transmitter knows what it sent. The receiver only observes what arrives.
How can the receiver reliably determine what was transmitted?

Attenuation



Signal loss over distance. E.g., The standard structured twisted-pair Ethernet channel limit is 100 meters. The signal doesn't vanish at 100m, but it degrades beyond reliable engineering performance standards.

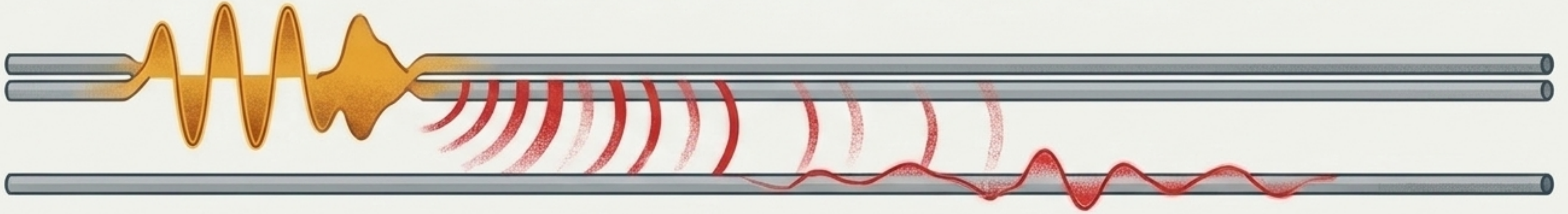
Noise



Unwanted external energy. The receiver does not operate in a perfectly silent environment.

Simply turning up transmit power doesn't fix this. You just end up amplifying the noise, too.

[The Problem - Near-End Crosstalk (NEXT)]



[The Solution - The Twist]

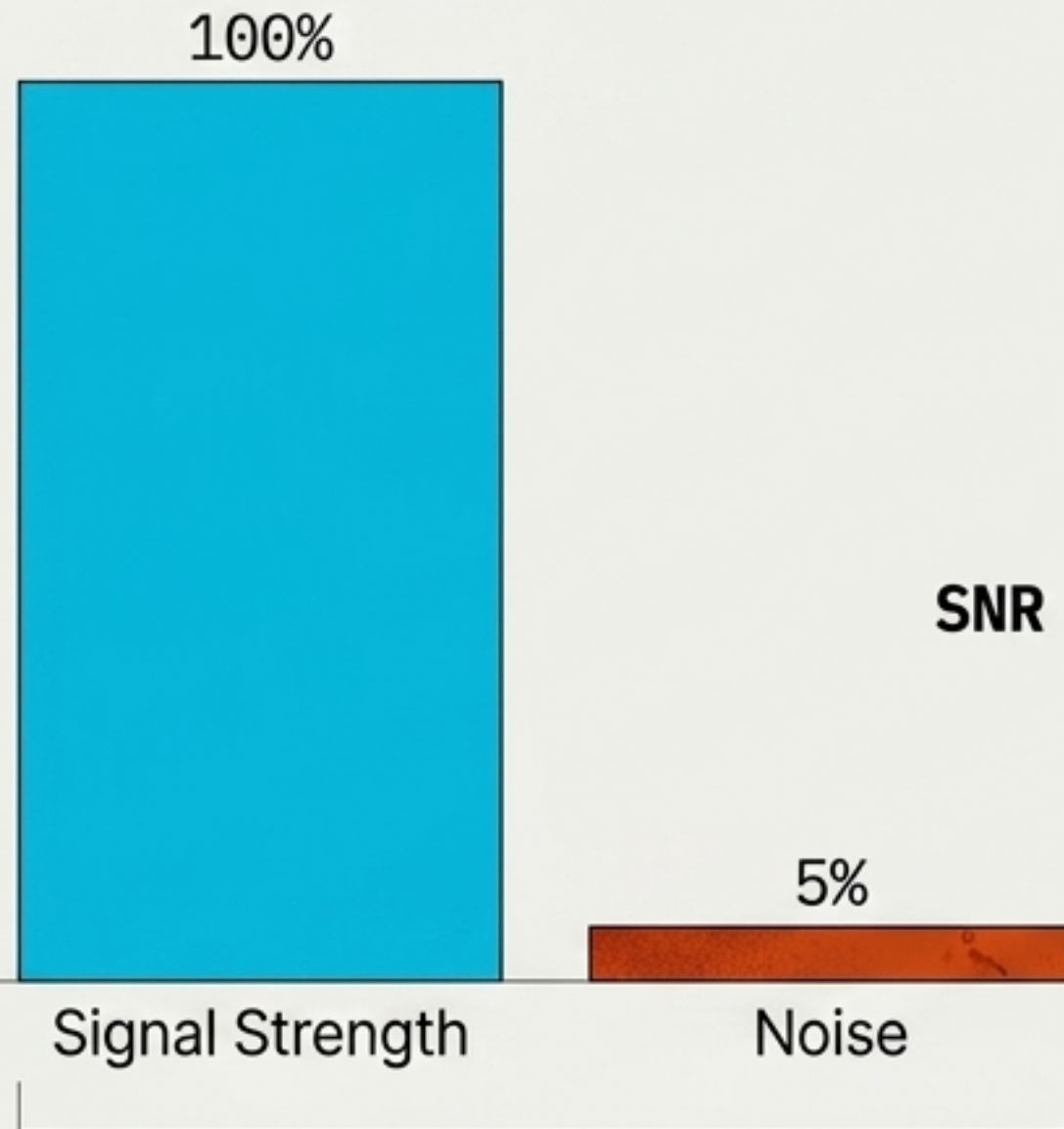


[Differential Signaling]

The twist is not decorative. By sending inverse signals down intertwined wires, the receiver measures the difference between the two. Any external noise hits both wires equally and is subtracted out, flatlining the interference.

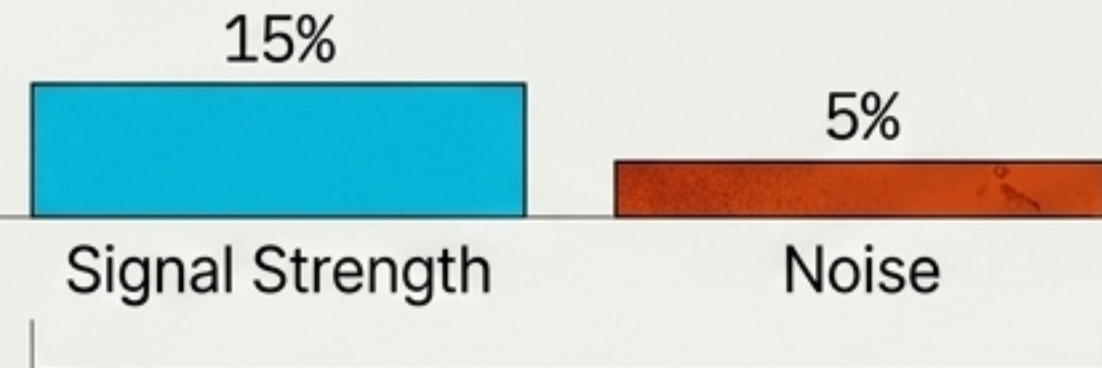
SIGNAL-TO-NOISE RATIO (SNR) CONCEPT

High SNR (Clear voice in a quiet room)



The receiver has massive separation to distinguish the intended data.

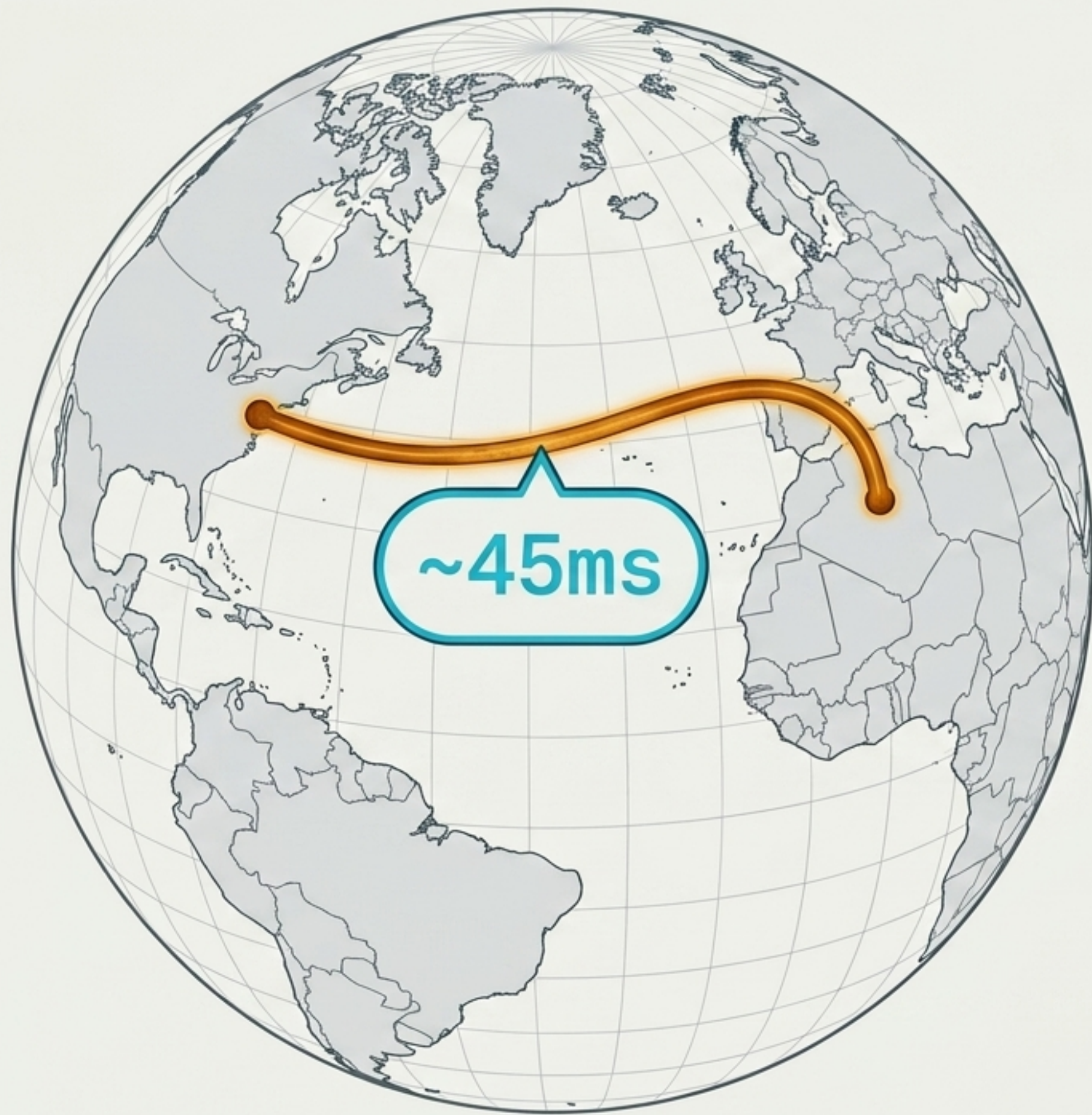
Low SNR (Talking in a loud factory)



As distance attenuates the signal, noise remains constant. The margin for error vanishes.

$$\text{SNR (dB)} = \frac{\text{Signal Strength}}{\text{Noise Strength}}$$

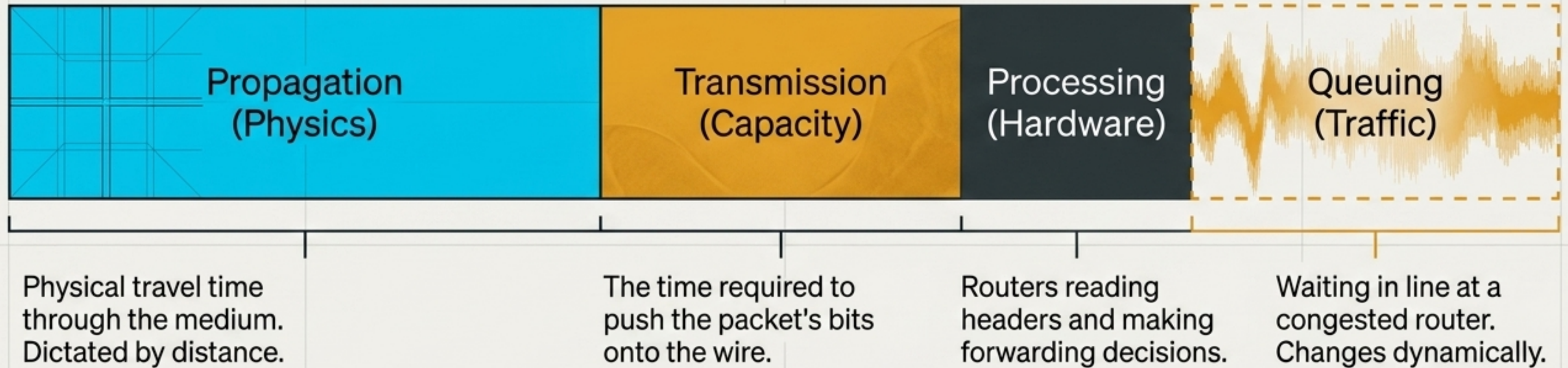
Increasing network speed requires **tighter signal integrity**. This is why **Cat5e**, **Cat6**, and **Cat8** exist—they aren't just '**faster**'; they are physically engineered for **superior crosstalk** and SNR performance.



Latency is a physics problem, not a software problem.

- The Speed of Light in a vacuum:
 3×10^8 m/s.
- In glass fiber or copper, it travels slower. Distances are vast. A perfectly engineered router and infinite bandwidth cannot outrun electromagnetic propagation.
- Some network problems are **configuration**. Some are technology. This one is **inescapable physics**.

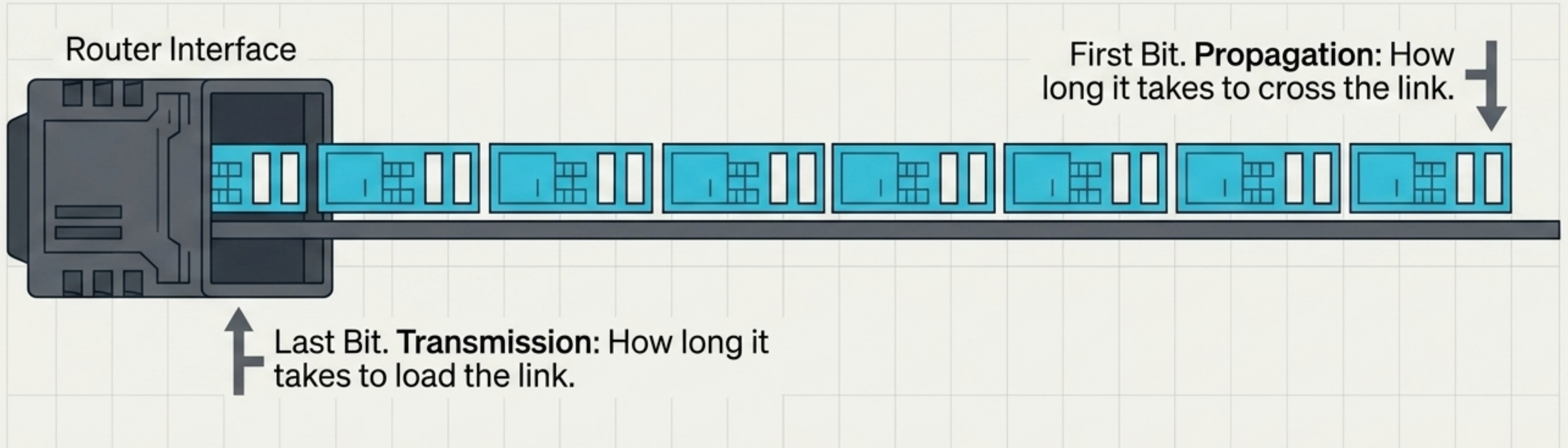
The Anatomy of RTT (Round-Trip Time)



Observed RTT = Propagation + Transmission + Processing + Queuing + (Return Path)

RTT is an observation of network behavior, not just a physical distance meter.

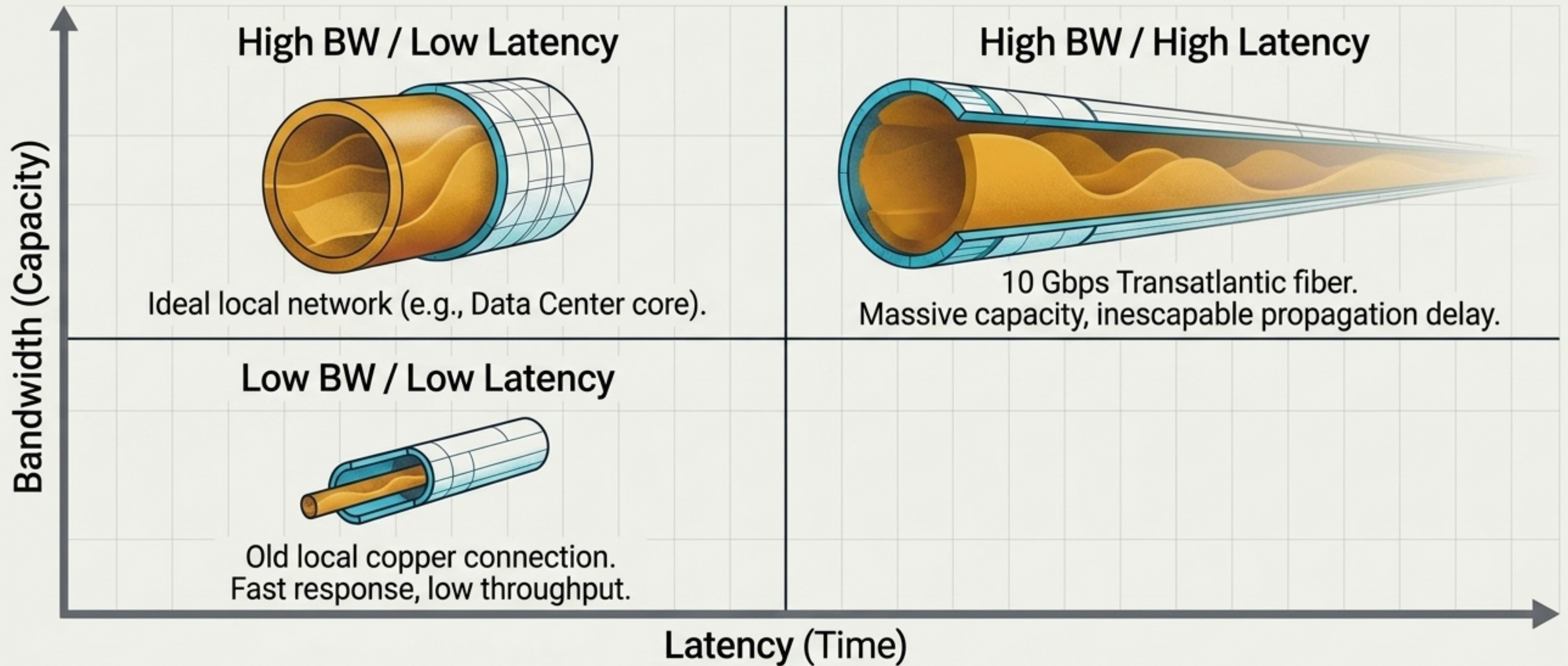
Propagation vs. Transmission



Example: A 1,000-byte packet (8,000 bits) on a 100 Mbps link.

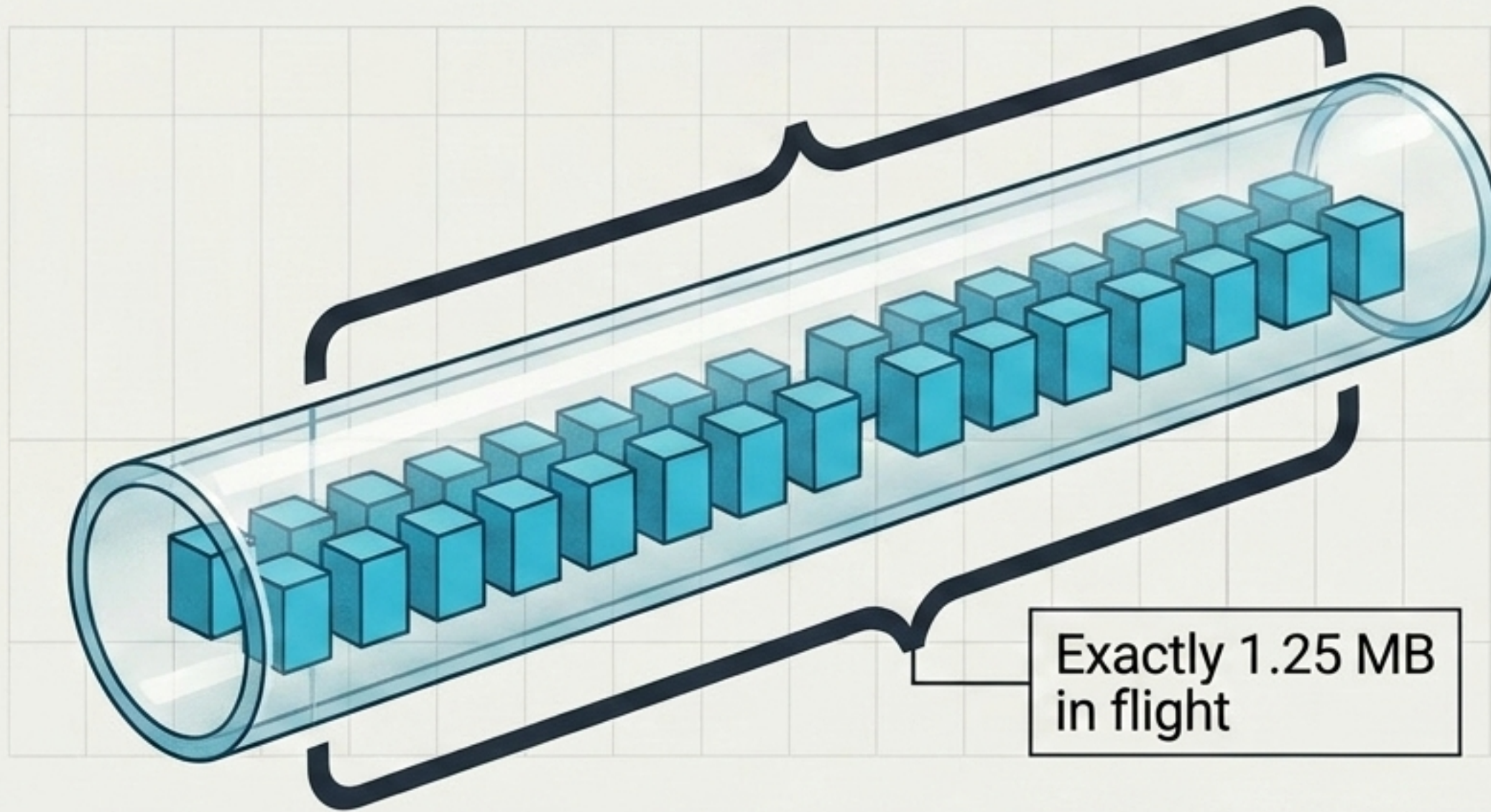
Transmission Delay = $8,000 / 100,000,000 = 0.08$ ms to push onto the wire.

Propagation Delay = Dependent entirely on the length of the cable.



Bandwidth is CAPACITY (how much fits). Latency is TIME (how long it takes to arrive).
Upgrading a 10 Gbps transatlantic link to 100 Gbps increases capacity, but the geographic distance remains identical. Interactive apps will still feel sluggish.

BDP: Data In Flight

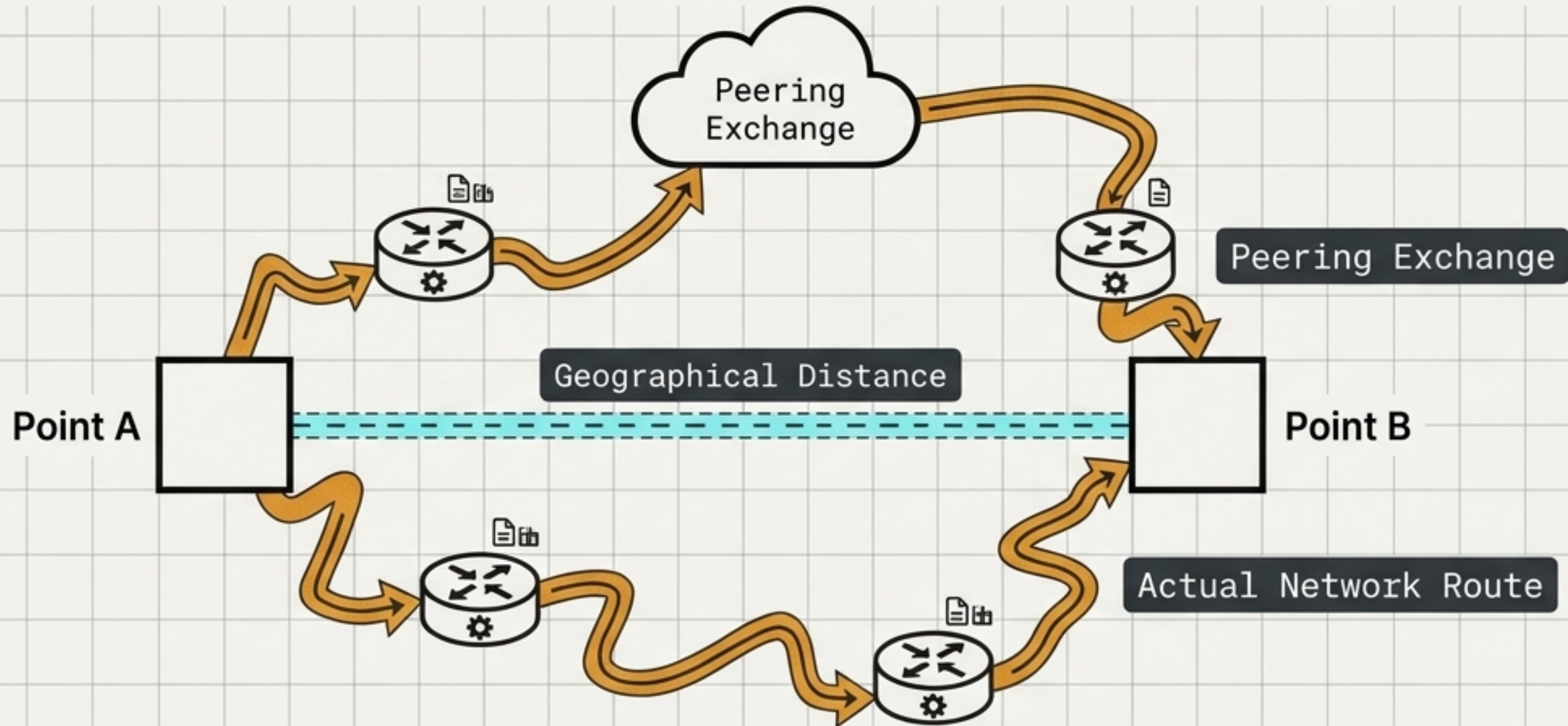


Bandwidth = 100 Mbps
RTT = 100 ms

$\text{BDP} = 100 \text{ Mbps} \times 0.1\text{s}$
 $= 10,000,000 \text{ bits}$
 $(\sim 1.25 \text{ MB})$

Data in Flight. On long-distance, high-capacity links, protocols like TCP must keep massive amounts of unacknowledged data physically "in flight" inside the wire just to fully utilize the available bandwidth.

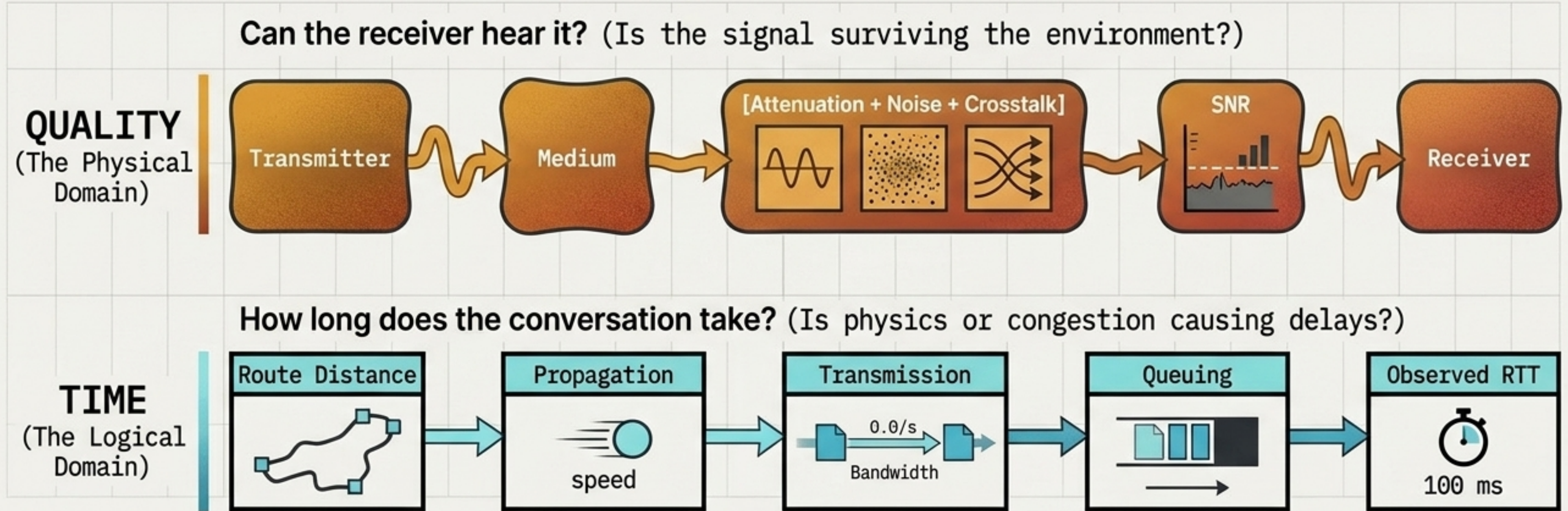
The Geography Trap (Measuring with Ping)



Ping measures the path, not just the geography. Geographically close destinations can have higher RTTs if the physical network route is convoluted, congested, or involves multiple carrier handoffs and queuing delays.

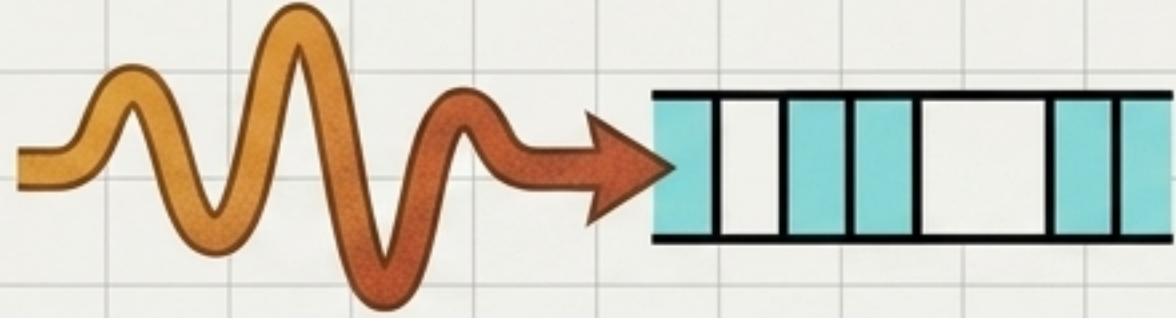
Packet Loss $\neq$ Latency.
They are different conditions. A destination can have low RTT but high packet loss, or zero packet loss with a massive RTT.

Synthesis: The Grand Mental Model



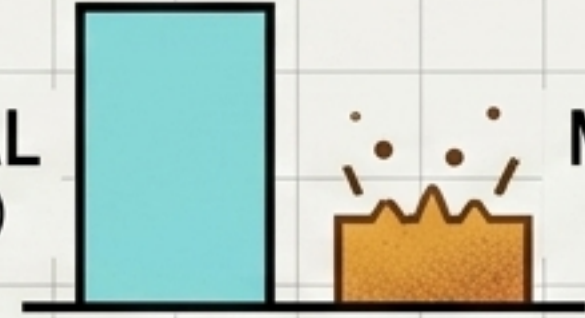
When a user says "The network is slow," it is rarely just "the internet." It is a failure of either Signal Quality or Physical Time. You cannot troubleshoot upper-layer protocols until you verify the foundation.

Key Takeaways



The physical layer bridges logic and reality.
Bits must become physical energy.

SIGNAL
(High)



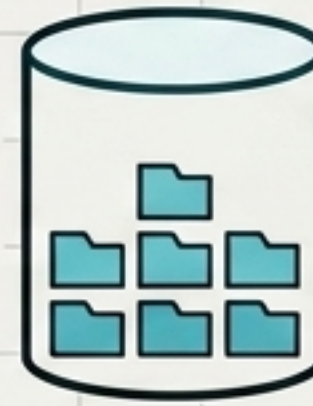
NOISE
(Low)

SNR dictates survival.
Attenuation and noise degrade signals;
twisted pairs and shielding protect them.



You cannot configure physics.
Propagation delay is an inescapable
constraint of geographic distance.

BANDWIDTH
(Capacity)



LATENCY
(Time)



Bandwidth $\neq$ Latency.
Bandwidth is capacity (how much fits).
Latency is time (how long it takes to arrive).